ANALYSTLAB AFRICA

Experience Lab Internship Programme

Official Weekly Assignment

Week 1: Airbnb Market Intelligence, Listing Pricing and Guest Experience Analytics Platform

Document Control

Programme	AnalystLab Africa Experience Lab
Week	Week 1
Project	Integrated Multidisciplinary Data Project
Project Title	Airbnb Market Intelligence, Listing Pricing and Guest Experience Analytics Platform
Version	1.0
Issue Date	_____
Submission Deadline	End of Week 1 (Thursday, 11:59 PM WAT)
Submission Method	Official AnalystLab Africa Week 1 Submission Form

1. PROJECT BACKGROUND

The short-term rental industry has experienced significant growth, with online accommodation platforms allowing property owners and hosts to offer different types of accommodation to travellers.

Airbnb provides a marketplace where users can explore accommodation options based on location, price, property type, room type and other listing characteristics.

For this project, AnalystLab Africa is simulating a professional accommodation-market analytics engagement using Airbnb listing and historical review records covering 10 major cities.

The supplied datasets provide information about property listings, host characteristics, advertised prices, geographical locations, room types, property attributes and historical review activity.

However, understanding this information requires more than simply counting listings or comparing advertised prices. The data must be properly inspected, cleaned, analysed and interpreted to identify meaningful market patterns.

The business requires a data-driven solution that can help stakeholders understand listing distribution, pricing patterns, property characteristics, host performance indicators and review activity across the selected cities.

As an intern at AnalystLab Africa, you have been assigned to an Airbnb Market Intelligence project team.

Your responsibility is to apply the skills acquired in your learning track to investigate the business problem and develop a solution that supports evidence-based market understanding and decision-making.

You are expected to demonstrate technical competence, analytical reasoning, appropriate methodology, data validation, professional documentation and an understanding of the limitations of the supplied data.

2. PROJECT PROBLEM STATEMENT

The accommodation marketplace operates across different cities, each with its own property characteristics, pricing patterns, accommodation types and market conditions.

Stakeholders require a better understanding of how Airbnb listings are distributed across cities, what property characteristics are associated with advertised prices, how host performance indicators differ and how historical review activity varies across listings.

Without structured analysis, it may be difficult to identify meaningful differences between markets or develop reliable insights that support property-market evaluation.

The project must investigate the following business questions:

1. Which cities have the highest and lowest numbers of Airbnb listings?
2. How do advertised listing prices vary across the 10 cities?
3. Which room types and property types are most common?
4. What listing characteristics are associated with higher advertised prices?
5. How do host characteristics and performance indicators vary across cities?
6. Which listings receive the highest review activity?
7. How do listing-level review scores differ across property types and locations?
8. Are there observable relationships between review activity and listing-level review ratings?
9. Can listing prices be estimated using property characteristics and other available listing information?
10. How can stakeholders access validated market insights through an interactive dashboard or AI-powered assistant?
11. How can the entire project be planned, coordinated, tested and delivered professionally?

Interns must answer these questions using the supplied data. They must not assume that the available information supports conclusions about actual occupancy, booking revenue, profitability or guest sentiment unless additional evidence is obtained.

3. PROJECT OBJECTIVES

The overall objective is to develop an integrated Airbnb Market Intelligence solution that helps stakeholders:

- Understand Airbnb listing distribution across the selected cities.
- Investigate advertised pricing patterns.
- Compare property types and room types.
- Examine host characteristics and available performance indicators.
- Analyse historical review activity.
- Investigate listing-level review ratings.
- Develop an appropriate price prediction model.
- Access validated market information through a Generative AI assistant.
- Coordinate the development and delivery of the overall solution through professional project-management practices.

By the end of the project, interns should demonstrate the ability to transform large datasets into meaningful insights and develop practical solutions using their respective technical disciplines.

4. DATASET DESCRIPTION

The project uses four supplied files.

Dataset	Description	Records/Fields
Listings.csv	Airbnb property listings, host information, prices, location and property characteristics	279,712 records, 33 columns
Reviews.csv	Historical review records associated with individual listings	5,373,143 records, 4 columns
Listings_data_dictionary(1).csv	Definitions of the fields in Listings.csv	33 field definitions
Reviews_data_dictionary(1).csv	Definitions of the fields in Reviews.csv	4 field definitions

4.1 Understanding the datasets

Listings.csv

This dataset contains information about individual Airbnb properties, including listing identifiers, host information, city, neighbourhood, advertised prices, room types, property characteristics and available listing-level review scores.

Reviews.csv

This dataset contains historical review records, including:

- `listing_id`
- `review_id`
- `date`
- `reviewer_id`

This file can be used to investigate review activity over time and the distribution of reviews across listings.

Important: The Reviews file does not contain the actual written review comments or individual review ratings. Therefore, interns cannot perform review-text sentiment analysis or summarise guest comments using this dataset.

4.2 Primary and foreign key relationships

The datasets contain a relationship that must be understood before analysis.

Listings table

- `listing_id` serves as the primary key.
- Each listing ID identifies an individual property record.

Reviews table

- `review_id` is an identifier for review records, but 160 repeated review IDs were detected in the supplied file.
- `listing_id` acts as the foreign key linking reviews to listings.
- The combination of `listing_id` and `review_id` is unique in the supplied records.

The relationship is one-to-many:

One listing can have multiple historical review records.

All review listing IDs in the supplied file match an existing listing ID.

However, not every listing has a corresponding review record in the supplied Reviews extract.

Interns must investigate the repeated review IDs and document their treatment rather than deleting records automatically.

4.3 Data preparation requirements

Before beginning their assigned tasks, all interns must:

1. Inspect all four supplied files.
2. Review the two data dictionaries.
3. Understand the structure and meaning of the available fields.
4. Check primary-key uniqueness.
5. Validate the listing-to-review relationship.
6. Investigate duplicate review identifiers.
7. Identify missing values.
8. Inspect numerical and categorical data types.
9. Investigate invalid or unusual advertised prices.
10. Document all cleaning and transformation decisions.

4.4 Important data limitations

A. Currency differences

Listing prices are expressed in the local currencies associated with their respective markets, and the supplied Listings table does not contain a dedicated currency field.

Therefore, interns must not directly combine or compare raw price amounts across cities as though they represent the same currency.

For example, an advertised price of 100 in New York cannot automatically be compared with 100 in Paris.

Interns should:

- Analyse prices separately within each city.
- Clearly state the currency context of each price comparison.
- Use documented and appropriately dated exchange rates if they choose to convert currencies.
- Explain any currency conversion assumptions.

B. Missing values

Some listing fields, particularly district information, host response indicators, host acceptance indicators, bedrooms and listing-level review scores, contain missing values.

Interns must investigate these missing values and select appropriate treatment methods.

C. Duplicate review identifiers

The Reviews file contains 160 duplicated `review_id` values. These must be investigated before treating `review_id` alone as a unique primary key.

D. Review information

The Reviews file contains review dates and identifiers but not the actual written review comments.

Therefore:

- Sentiment analysis using actual review text is not supported.
- Guest comment summarisation is not supported.
- Review frequency and historical review activity analysis are supported.
- Listing-level review score analysis is supported where those scores are available in Listings.csv.

E. Occupancy and revenue limitations

The supplied datasets do not establish actual bookings, completed stays, occupancy rates or realised rental revenue.

Listing prices must not be presented as actual revenue or profit.

All findings must reflect the information available in the supplied records.

5. COMMON REQUIREMENTS FOR ALL INTERNS

Regardless of your assigned track, every intern must complete the following foundational activities.

Task 1: Understand the business problem

Write a brief explanation of the Airbnb market intelligence challenge.

Your explanation must identify:

- The business problem you intend to investigate.
- Why the problem matters to the accommodation marketplace.
- The intended user of your solution.
- The business decisions your findings or solution could support.
- The limitations of the available evidence.

Task 2: Inspect the datasets

Review the relevant datasets and document:

- The files and columns you intend to use.
- The number of records and columns.
- Candidate primary keys and foreign keys.
- Missing values and duplicate records.
- Inconsistent data types.
- Price and currency considerations.
- The relationship between Listings and Reviews.
- Limitations that may affect your analysis.

Task 3: Define success criteria

Identify at least three measurable criteria that will help determine whether your work has achieved its objectives.

Examples include:

- Analytical calculations reconcile with independent checks.
- A dashboard displays accurate and consistent measures.
- A machine-learning model outperforms an appropriate baseline.
- An AI assistant produces supported answers to a defined test set.
- A project plan contains clear milestones, owners, dependencies and acceptance criteria.

Your success criteria must relate to your assigned track.

Task 4: Maintain professional standards

- Use clear file names and organised project folders.
- Document assumptions and important decisions.
- Attribute external sources and tools used.
- Include evidence of testing and validation.
- Do not expose unnecessary host or reviewer identifying information.
- Maintain reproducible workflows.
- Do not present unsupported conclusions as established facts.

6. TRACK-SPECIFIC ASSIGNMENTS

TRACK A — DATA ANALYTICS

Assignment Title: Airbnb Market Performance and Listing Insights Dashboard

Objective

Develop an interactive dashboard that enables stakeholders to understand Airbnb listing distribution, advertised pricing, property characteristics, host indicators and historical review activity across the 10 selected cities.

Task 1: Prepare and model the data

1. Import `Listings.csv` into your chosen analytical environment.
2. Load `Reviews.csv` into an appropriate database or data-processing environment.
3. Inspect the data types and available columns.
4. Assess missing values, duplicates and unusual values.
5. Validate `listing_id` in the Listings table.
6. Investigate duplicated `review_id` values.
7. Establish and validate the listing-to-review relationship.
8. Create an appropriate data model.
9. Document transformations and explain why they were necessary.

Task 2: Analyse Airbnb market performance

Investigate at least six of the following questions:

1. Which cities have the highest number of listings?
2. How are listings distributed across room types?
3. Which property types are most common?
4. How do advertised prices vary across cities, within each local currency?
5. Which neighbourhoods or districts have the largest listing concentrations, where the data is sufficiently complete?
6. Which listings have the highest available review scores?
7. How many reviews are recorded for each listing?
8. How does review activity change over time?
9. How do host response and acceptance indicators vary across cities?

10. Which property characteristics are most common among higher-priced listings?

For every question, identify the relevant KPI, calculation method and business interpretation.

Task 3: Develop the dashboard

Your dashboard should contain the following sections:

Page 1: Airbnb Market Overview

- Total listings.
- Number of cities represented.
- Listing distribution by city.
- Room type distribution.
- Property type distribution.

Page 2: Pricing and Property Analysis

- Price distribution by city.
- Price comparison within local currencies.
- Price variation by room type.
- Price analysis by accommodation capacity or other supported property characteristics.

Page 3: Host Performance Analysis

- Listings per host.
- Host response indicators.
- Host acceptance indicators.
- Available host characteristics and comparisons.

Page 4: Review Activity and Listing Scores

- Total review records.
- Review activity over time.
- Review volume by listing or city.
- Listing-level review score comparisons.
- Review activity versus available aggregate listing ratings.

Include filters for relevant cities, room types, property categories and other supported fields.

Task 4: Validate the dashboard

- Independently verify at least three important calculations.
- Check whether totals respond correctly to filters.
- Validate listing and review counts.
- Investigate discrepancies between source records and dashboard results.
- Document limitations that could affect interpretation.

Task 5: Produce business insights

Provide at least five meaningful findings and three actionable recommendations.

Each recommendation must identify the evidence supporting it, the proposed action and how stakeholders could evaluate its usefulness.

Required Deliverables

1. Power BI `.pbix` file.
2. SQL queries or Python data preparation scripts.
3. Data-quality assessment.
4. KPI definition sheet.
5. Dashboard screenshots.
6. Validation evidence.
7. Executive summary of findings and recommendations.
8. README documentation.

Recommended tools: SQL, PostgreSQL or DuckDB, Python, pandas or Polars, Power BI Desktop, Power Query and DAX.

TRACK B — DATA SCIENCE

Assignment Title: Airbnb Pricing, Location and Listing Characteristics Investigation

Objective

Investigate how listing characteristics, room types, location, host attributes and review-related indicators vary across the selected Airbnb markets.

The goal is to use statistical analysis to discover patterns and relationships that can support better understanding of the accommodation marketplace.

Task 1: Define your research questions

Select one primary research problem and at least two supporting research questions.

Possible questions include:

- How does advertised listing price vary across room types within individual cities?
- Are the number of bedrooms and accommodation capacity associated with advertised prices?
- Do host response indicators differ across cities?
- Are listing-level review scores associated with differences in room or property types?
- Is review activity associated with available listing-level review ratings?
- How does listing distribution vary across neighbourhoods or districts?

Clearly state:

- Your research problem.
- Why it matters.
- The variables required.
- The expected analytical outcome.
- The limitations of the available data.

Task 2: Conduct exploratory data analysis

1. Inspect the relevant datasets.
2. Analyse the distribution of prices within each city.
3. Examine room types and property categories.
4. Investigate missing values and unusual price values.
5. Examine listing-level review scores.
6. Analyse review activity by listing and over time.
7. Investigate relationships between selected variables.
8. Create appropriate visualisations.
9. Explain what the results demonstrate and what they cannot establish.

Task 3: Formulate and test hypotheses

Develop at least three hypotheses relevant to your selected research problem.

For each hypothesis:

- State the null and alternative hypotheses where applicable.
- Identify the variables.
- Select an appropriate statistical test.
- Explain the assumptions of the selected method.
- Conduct the analysis.
- Interpret the results.
- Discuss statistical and practical significance.
- Identify limitations.

Examples include:

- Advertised prices differ across room types within a selected city.
- Accommodation capacity is associated with advertised listing prices.
- Review activity differs across listing categories.
- Aggregate review ratings differ across selected property groups.

Do not compare raw prices across different currencies without appropriate currency standardisation.

Task 4: Investigate limitations

Explain how the following could affect your findings:

- Missing listing attributes.

- Currency differences.
- Unequal listing counts across cities.
- Duplicate review identifiers.
- Different levels of review activity.
- Missing district information.
- Potentially influential extreme prices.
- The absence of actual booking and occupancy records.

Distinguish between association and causation.

Task 5: Produce an analytical report

Your report must contain:

1. Research problem and objectives.
2. Dataset inspection and preparation.
3. Exploratory data analysis.
4. Research questions and hypotheses.
5. Statistical methods and assumptions.
6. Results and interpretations.
7. Visualisations.
8. Limitations.
9. Recommendations.

Required Deliverables

1. Reproducible Python notebook.
2. Data-quality summary.
3. At least five relevant visualisations.
4. Statistical analysis report.
5. Research findings and recommendations.
6. Instructions for reproducing the analysis.

Recommended tools: Python, pandas, NumPy, Matplotlib, Seaborn, SciPy and Jupyter Notebook.

TRACK C — MACHINE LEARNING

Assignment Title: Airbnb Listing Price Prediction Model

Objective

Develop and evaluate a machine-learning model that estimates advertised Airbnb listing prices using available property characteristics, host information and other relevant listing attributes.

The model should demonstrate how machine learning can support market analysis and listing-price estimation.

It must not be presented as a guaranteed pricing recommendation or actual revenue prediction.

Task 1: Define the prediction problem

Your primary assignment is to predict the advertised price of an Airbnb listing using relevant listing characteristics.

Investigate whether the available variables can support an appropriate prediction target.

Potential features include:

- City or location.
- Room type.
- Property type.
- Accommodation capacity.
- Number of bedrooms.
- Number of bathrooms, where available.
- Host response indicators.
- Available listing-level review scores.

Select features that are available and appropriate at the intended prediction point.

Task 2: Prepare the modelling dataset

1. Inspect the price field and determine its format.
2. Remove or appropriately handle invalid and non-positive price values.
3. Investigate extreme price observations.
4. Handle missing values appropriately.
5. Encode categorical variables.
6. Perform feature selection and engineering.
7. Prevent target leakage.
8. Document the preprocessing pipeline.

Currency requirement:

Do not train one pooled price model on raw prices from different currencies without appropriate conversion.

You may choose one of two approaches:

- Build separate city-specific models using each city's local currency.
- Develop a standardised-price model using documented exchange rates and clearly stated assumptions.

Explain the approach selected.

Task 3: Establish a baseline

1. Create a simple baseline prediction.

2. Select appropriate evaluation metrics.
3. Split the dataset into training and testing sets.
4. Explain why the selected split is appropriate.
5. Establish baseline performance before developing more complex models.

Task 4: Train and compare models

Train and compare at least two suitable machine-learning approaches.

Possible models include:

- Linear Regression.
- Decision Tree Regressor.
- Random Forest Regressor.
- Gradient Boosting Regressor.
- Other justified regression methods.

Evaluate the models using appropriate metrics, including:

- Mean Absolute Error (MAE).
- Root Mean Squared Error (RMSE).
- R^2 score.

Discuss why the chosen metrics are appropriate and interpret the results within the relevant city and currency context.

Task 5: Test and package the solution

- Build a reproducible preprocessing and training workflow.
- Evaluate prediction errors.
- Compare model performance.
- Investigate overfitting and underfitting.
- Include appropriate input validation.
- Document how the model can be loaded and used.
- Explain the limitations of the model.
- Identify additional information that would improve future model performance.

Required Deliverables

1. Reproducible Python notebook or source code.
2. Data preparation and feature engineering documentation.
3. Target and feature definitions.
4. Baseline and model comparison report.
5. Evaluation metrics and error analysis.
6. Feature importance or model interpretation where appropriate.
7. Saved model.
8. Inference script or working prediction prototype.
9. Tests and README.

Recommended tools: Python, pandas, NumPy, scikit-learn, Matplotlib, Git and Jupyter Notebook.

TRACK D — GENERATIVE AI

Assignment Title: Airbnb Market and Listing Insights Assistant

Objective

Build an AI-powered assistant that helps users understand Airbnb market information, compare available listing attributes and retrieve validated insights from the supplied datasets.

The assistant should provide natural-language access to structured listing information and review activity.

Task 1: Define the use case

Identify the intended users of the assistant and at least 10 questions it should answer.

Example questions:

- Which city has the largest number of Airbnb listings?
- Which room types are most common in Paris?
- What is the median advertised listing price in New York?
- Which property types are most common in Sydney?
- How many review records are associated with a particular listing?
- Which listings have the highest available aggregate review ratings?
- How does review activity vary across cities?
- What listing characteristics are available for price comparison?
- What does the host response rate field mean?
- What limitations should users understand when interpreting the dataset?

The assistant must identify when a question cannot be answered using the available information.

Task 2: Prepare the knowledge sources

1. Review the Listings data dictionary.
2. Review the Reviews data dictionary.
3. Identify the approved listing fields and definitions.
4. Prepare validated summaries and analytical outputs.
5. Identify how numerical queries will be calculated.
6. Document missing information and limitations.
7. Prepare suitable data sources for the assistant.

The Reviews file does not contain written review comments. Do not build a text-sentiment analysis or guest-comment summarisation feature using this dataset.

Task 3: Build the application

Create a working AI application that:

- Accepts natural-language questions.
- Uses approved listing information and validated review aggregates.
- Retrieves or calculates the relevant information.
- Produces understandable responses.
- Provides supporting figures or source references where feasible.
- Handles unsupported questions appropriately.
- Explains currency context when answering price questions.
- Does not expose unnecessary host or reviewer information.

For numerical questions, use validated SQL queries or controlled analytical functions rather than asking the language model to invent figures.

Task 4: Evaluate the assistant

Create a test set of at least 20 questions covering:

- Directly supported questions.
- Numerical calculations.
- Listing comparisons.
- Review activity.
- Ambiguous questions.
- Questions with missing information.
- Currency comparisons requiring clarification.
- Unsupported requests, such as actual occupancy or booking revenue.
- Requests for unavailable review-text sentiment analysis.

For each test question, record:

- The question.
- Expected answer or expected behaviour.
- Actual answer.
- Whether the answer is supported.
- Accuracy result.
- Corrective action, where necessary.

Task 5: Improve reliability

Document how you addressed:

- Incorrect answers.
- Unsupported responses.
- Incorrect numerical results.
- Currency interpretation issues.
- Privacy considerations.
- Failure cases.

- Retrieval or query errors.

Required Deliverables

1. Working AI assistant prototype.
2. Source code and setup instructions.
3. Description of the data sources and system architecture.
4. System instructions and configuration.
5. At least 20 evaluation questions.
6. Evaluation report.
7. Responsible AI and data-protection assessment.
8. Demonstration video or screenshots.
9. README documentation.

Recommended tools: Python, an appropriate LLM API, Streamlit or Gradio, SQL, and a retrieval or structured-query approach.

TRACK E — PROJECT MANAGEMENT

Assignment Title: Airbnb Market Intelligence Project — Planning, Governance and Delivery Control

Objective

Prepare an execution-ready project management plan for the delivery of the Airbnb Market Intelligence Platform.

Your plan must account for the different technical workstreams, dependencies, data-quality considerations, technical requirements and delivery constraints.

Task 1: Develop the project charter

Prepare a project charter containing:

- Project background and business need.
- Project objectives.
- Scope and exclusions.
- Expected deliverables.
- Stakeholders and responsibilities.
- Assumptions and constraints.
- Success criteria and acceptance requirements.
- Major project risks.

Task 2: Create the Work Breakdown Structure

Develop a structured WBS covering:

- Project initiation.

- Dataset inspection.
- Data preparation.
- Data analytics.
- Data science.
- Machine-learning development.
- Generative AI development.
- Testing and validation.
- Integration.
- Documentation.
- Final presentation and handover.

Identify the main deliverables, activities and dependencies.

Task 3: Develop the project schedule

Prepare a Gantt chart containing:

- Project activities.
- Task owners.
- Planned start and completion dates.
- Estimated durations.
- Milestones.
- Dependencies.
- Progress tracking.
- Critical-path analysis where appropriate.

Explain your assumptions when estimating the schedule.

Task 4: Prepare the resource and budget plan

Identify the resources required to deliver the project.

Include relevant categories such as:

- Human resources.
- Software and tools.
- Database hosting.
- Computing resources.
- Development and testing.
- Documentation.
- Contingency.

Clearly label all figures as estimates and document the assumptions behind them. Do not present hypothetical costs as actual business expenditure.

Task 5: Develop a risk and quality management plan

Prepare a risk register containing at least eight project risks.

Consider:

- Missing listing information.
- Duplicate review identifiers.
- Large review dataset processing requirements.
- Currency differences.
- Incorrect dashboard calculations.
- Machine-learning model errors.
- AI-generated inaccurate responses.
- Delayed track deliverables.
- Integration problems.
- Data privacy and unauthorised disclosure.

For each risk, document:

- Risk description.
- Likelihood.
- Impact.
- Mitigation strategy.
- Responsible owner.
- Current status.

Also define quality gates and acceptance criteria for the major project deliverables.

Task 6: Simulate project progress

Create a hypothetical project status scenario in which:

- At least two activities are delayed.
- One important dependency is affected.
- One change request is raised.

Explain the impact on project scope, schedule, resources and deliverable quality.

Recommend corrective actions and update the relevant project records.

Clearly identify the scenario as a simulation rather than an actual business event.

Required Deliverables

1. Project charter.
2. Work Breakdown Structure.
3. Gantt chart and dependency analysis.
4. Resource and estimated budget plan.
5. Risk register.
6. Quality and acceptance plan.
7. Stakeholder communication plan.
8. Simulated progress report.
9. Change log.

10. Final project delivery and handover checklist.

Recommended tools: Microsoft Project, Trello, Jira, Asana, Notion or an equivalent project-management platform. Microsoft Excel is not permitted.

7. SHARED INTEGRATION AND COLLABORATION REQUIREMENTS

All five tracks contribute to the same overall Airbnb Market Intelligence Platform.

However, each intern remains responsible for completing the technical deliverables of their assigned track.

The team must establish consistent definitions for key concepts, including:

- Listing count.
- Advertised price.
- Room type.
- Property type.
- Host performance indicators.
- Review count.
- Review activity.
- Listing-level review ratings.

The project should bring together the following components where technically appropriate:

1. Market Performance Dashboard

Validated listing counts, pricing comparisons, city-level analysis and review activity.

2. Statistical Analysis

Evidence-based findings on listing characteristics, price differences and review-related patterns.

3. Price Prediction Prototype

A model that estimates advertised listing prices using suitable property characteristics and correctly handled currency information.

4. AI-Powered Market Insights Assistant

A natural-language interface that retrieves and explains validated listing information and analytical findings.

5. Project Management Documentation

Project scope, milestones, risks, testing, progress tracking and final handover.

Interns must provide evidence of how their individual work supports the overall project.

Do not claim integration if an output has not actually used or consumed another track's work.

8. TECHNICAL TOOLS AND NO-EXCEL POLICY

Mandatory requirement

Microsoft Excel is not permitted for this assignment.

Because the Reviews dataset contains more than five million records, interns must use appropriate data-processing and analytical tools.

The recommended tools are:

Track	Approved tools
Data Analytics	SQL, PostgreSQL, DuckDB, Python, Power BI Desktop
Data Science	Python, pandas, Polars, NumPy, SciPy, Jupyter Notebook
Machine Learning	Python, pandas, scikit-learn and appropriate modelling libraries
Generative AI	Python, LLM APIs, Streamlit, Gradio and SQL
Project Management	Microsoft Project, Trello, Jira, Asana or Notion

Data processing requirements

- Use SQL or Python to load and process the large Reviews dataset.
- Do not attempt to load the entire Reviews.csv directly into a spreadsheet.
- Use database queries or Python aggregation to produce summaries where needed.
- Use Power BI for reporting and visualisation.
- Preserve original datasets and create separate working or transformed datasets.
- Document processing decisions and validation checks.
- Keep code and outputs reproducible.

9. FINAL SUBMISSION REQUIREMENTS

Each intern must submit the following:

1. Track-specific technical deliverables and working files.
2. Source code, SQL queries, dashboard or project artefacts, as appropriate.
3. A README explaining the project purpose, tools, setup instructions, methodology, assumptions and limitations.
4. Evidence of data inspection, validation and testing.
5. Findings, recommendations or technical solutions relevant to the assigned track.
6. Evidence of individual contribution and collaboration.
7. A record of meaningful improvements made after receiving feedback.
8. An individual recorded presentation of approximately 5–10 minutes explaining the work, results, limitations and contribution to the overall project.

All files must be properly organised using clear file names and folder structures.

A reviewer should be able to understand the work and reproduce the relevant analysis or development process where applicable.

10. ASSESSMENT AND QUALITY EXPECTATIONS

Interns will be assessed according to the requirements of their assigned track.

Assessment will consider:

- Understanding of the business problem.
- Dataset inspection and quality.
- Technical execution and methodology.
- Testing and validation.
- Accuracy and usefulness of findings or solutions.
- Business interpretation and recommendations.
- Integration and collaboration.
- Documentation and reproducibility.
- Professional communication.

The technical expectations must reflect the responsibilities of each discipline.

A Data Analytics intern will be assessed on dashboard quality and KPI accuracy, while a Machine Learning intern should be assessed on model development, evaluation and error analysis.

A Generative AI intern will be assessed on response grounding, accuracy and responsible AI implementation, while a Project Management intern should be evaluated on planning, coordination, risk management and delivery evidence.

Each track will use its own dedicated 100-mark grading rubric.

11. DEFINITION OF SUCCESS

The project will be considered successful when interns deliver technically sound, documented and validated solutions that demonstrate how Airbnb listing and review data can support market understanding and informed business decisions.

The completed solution should provide:

- Reliable reporting of listing distribution.
- Meaningful analysis of advertised pricing.
- Comparisons of property and room types.
- Insights into host characteristics and available performance indicators.
- Analysis of historical review activity.
- A defensible listing-price prediction prototype.
- An AI-powered interface grounded in validated information.
- Professional project planning, coordination and delivery documentation.

All conclusions must be supported by the supplied records.

Interns must distinguish advertised prices from actual rental revenue, review activity from customer satisfaction, and statistical associations from causal explanations.

The overall project should demonstrate professional problem-solving, responsible data use, reproducible work, effective communication and multidisciplinary collaboration.

PROJECT SOURCE AND ATTRIBUTION

Dataset: Airbnb Listings and Reviews Dataset.

Supplied Files:

1. Listings.csv
2. Reviews.csv
3. Listings_data_dictionary(1).csv
4. Reviews_data_dictionary(1).csv

Professional Development Requirement

As part of building your professional portfolio:

• Update your GitHub repository with your Week 2 project.

• Publish a LinkedIn post describing your project

• Publish an X (Twitter) post summarizing your project.

• Tag AnalystLab Africa on LinkedIn.

• Tag @analystlabafric on X.

• Include the hashtag #AnalystLabAfrica on both platforms.

Professional Branding Reminder

AnalystLab Africa encourages every intern to document their learning journey professionally on LinkedIn and X. Consistently sharing your work helps build your portfolio and increases your visibility to recruiters and employers.